

PROFESSIONAL SQL CASE STUDY / 2021-2025

US Domestic Passenger Traffic Analysis

Scheduled segment demand, carrier capacity and airport activity

Analysis rows	Years	Months per year	2025 passengers
1,579,887	5	12	845.17M

Project purpose

Use Google BigQuery and the Bureau of Transportation Statistics (BTS) T-100 Domestic Segment data to explain how scheduled U.S. carrier passenger traffic changed between 2021 and 2025. The workflow checks five annual extracts, defines one consistent analysis population, and answers six practical questions with transparent aggregates.

Platform / Google BigQuery Sandbox

Primary source / BTS TranStats T-100 Domestic Segment (U.S. carriers)

Analysis period / Complete calendar years 2021-2025

01 / EXECUTIVE SUMMARY

Scheduled segment passengers rose from 610.99 million in 2021 to 856.42 million in 2024, then fell to 845.17 million in 2025. The 2025 decline was about 11.25 million passengers (-1.31%) as seats rose by about 11.99 million (+1.14%). The distance-weighted load factor dropped from 83.71% to 81.70%.

Key results and decision value

Business question	Primary finding	Decision value
1 Annual trend	2025 demand eased while capacity grew	Monitor utilization with demand
2 Monthly pattern	Nine of 12 months fell versus 2024	Prioritize monthly route review
3 Carrier shifts	Spirit had the largest loss; United the largest gain	Investigate carrier-level mix
4 Top routes	LAX to JFK led directional segments	Context for route investigation
5 Carrier capacity	Southwest supplied the most seats	Compare supply with route utilization
6 Origin airports	Atlanta led departing segment passengers	Context for airport investigation

Business objective

Describe five-year scheduled passenger demand and capacity, locate the 2025 change by month and reporting carrier, and identify leading directional segments and departure airports. Every comparison uses the same filtered analysis view.

Business questions

1. How did passengers, seats, departures and utilization change annually?
2. Which months drove the 2024-to-2025 change?
3. Which airlines gained or lost the most passengers between 2024 and 2025?
4. Which directional nonstop segments carried the most passengers in 2025?
5. Which carriers supplied the most seats in 2025?
6. Which airports had the most departing segment passengers in 2025?

02 / SOURCE, SCOPE AND SQL WORKFLOW

Source and analytical grain

BTS TranStats T-100 Domestic Segment (U.S. carriers) was downloaded as five annual CSV extracts. Each row aggregates one month of performed nonstop segments at a carrier, origin, destination, aircraft type and service-class combination. Rows do not identify individual flights or unique travelers. Airport IDs and codes are retained; display names in the results expand the codes for readers.

Scope element	Definition
Period	January 2021 through December 2025; 12 months present in each year
Service	Class F scheduled passenger/cargo service in the BTS field
Source files	Five yearly BTS T-100 Domestic Segment (U.S. carriers) CSVs
BigQuery source	portfolio-project-508812.us_domestic_passenger_traffic.t100_domestic_segments
Analysis object	vw_scheduled_segments_2021_2025, a filtered reusable view

Five-year ingestion and analysis coverage

Year	Raw rows	Analysis rows	Months	Analysis passengers
2021	381,352	270,662	12	610,993,050
2022	408,432	297,562	12	757,182,049
2023	424,178	318,370	12	825,911,334
2024	437,871	338,164	12	856,422,445
2025	455,857	355,129	12	845,169,134
Total	2,107,690	1,579,887	-	3,895,678,012

Raw rows include all source service classes. Analysis rows and passengers are from the class F view after validity rules; they must not be compared as if they were the same population.

Workflow

Stage	Purpose	Output
1 Load	Apply the same schema to each annual CSV	Combined source table
2 Profile	Check service classes and suspicious values	Queries 01 and 02
3 Prepare	Define one consistent valid scheduled population	View in query 03
4 Analyze	Answer the six business questions	Queries 04-09
5 Interpret	Document evidence, limits and next data needs	This case study

03 / DATA VALIDATION AND QUALITY

A single reusable analysis view

For all six questions we use the same saved selection of records, called `vw_scheduled_segments_2021_2025`. A row is one monthly summary for a carrier, route, aircraft type and service class - not one flight or one person. We select Class F scheduled service from 2021-2025, then check that month, departures, seats, passenger count and distance are usable.

How we checked the Class F records

Imagine reviewing forms before calculating a total. We began with 1,592,187 Class F records. The table shows how many had each condition and whether that condition removes the record.

What we checked	Records	What happens
All Class F records	1,592,187	Starting set
Zero passengers	60,358	Keep if other checks pass
Departures zero, negative or blank	3,172	Leave out
Seats zero, negative or blank	3,812	Leave out
Negative or missing passengers	0	None found
More passengers than seats	111	Leave out
Distance zero, negative or blank	8,955	Leave out
Month blank or outside 1-12	0	None found

What the result means: 1,579,887 records passed the checks; 12,300 of the starting Class F records were left out. A zero passenger count alone does not remove a record.

Example: A monthly record with 10 departures, 500 seats, 0 passengers, a valid month and a 100-mile distance stays in the analysis. If that same record has 0 seats, it is left out. Keeping a valid zero-passenger record preserves its offered seats when we compare passengers with capacity.

Why the problem counts do not add up to 12,300: The same record may have both zero seats and zero departures. It appears in both checks above but is left out only once. The 60,358 zero-passenger count is from the starting data; some of those records may fail another check.

Fields and meanings

Field	Meaning in this report
YEAR, MONTH	Reporting period; every year has 12 observed months.
UNIQUE_CARRIER / NAME	Reporting carrier code and label; regional operators can appear separately.
ORIGIN, DEST	Airport codes for the direction of a nonstop segment.
DEPARTURES_PERFORMED	Number of performed departures represented by the row.
PASSENGERS, SEATS	Segment passengers and available segment seats.
DISTANCE	Segment distance in miles; used as a weight for passenger/seat miles.
CLASS, AIRCRAFT_TYPE	Service class and aircraft-type code; aircraft codes retained as text.

04 / FINDINGS - DEMAND AND UTILIZATION

Q1. How did demand, capacity and utilization change from 2021 to 2025?

Year	Passengers (M)	Seats (M)	Departures	Distance-weighted LF
2021	610.99	803.03	6,754,231	77.68%
2022	757.18	925.16	7,412,094	84.32%
2023	825.91	1,018.51	7,797,919	83.53%
2024	856.42	1,053.84	8,125,082	83.71%
2025	845.17	1,065.83	8,322,290	81.70%

Finding:

Passengers increased each year from 2021 through 2024, then fell by about 11.25 million (-1.31%) in 2025. Seats rose by about 11.99 million (+1.14%) and departures by 197,208, while distance-weighted load factor fell 2.01 percentage points to 81.70%. Thus more service was offered against slightly fewer recorded segment passengers; the table cannot explain whether fares, schedules, competition or other factors caused that pattern.

04 / FINDINGS - MONTHLY DEMAND

Q2. Which 2025 months drove the change from 2024?

Month	2024 pax (M)	2025 pax (M)	Change	2024 seats (M)	2025 seats (M)
Jan	60.28	60.60	+0.52%	79.31	80.71
Feb	61.36	58.43	-4.77%	77.91	76.99
Mar	73.59	72.56	-1.39%	89.13	92.14
Apr	70.90	69.90	-1.41%	87.16	89.47
May	76.28	74.59	-2.23%	91.08	92.51
Jun	77.64	76.76	-1.13%	91.73	93.12
Jul	79.31	79.39	+0.11%	93.86	95.66
Aug	75.22	74.89	-0.43%	91.80	92.00
Sep	68.25	67.64	-0.89%	86.42	85.66
Oct	73.29	74.54	+1.71%	91.09	92.64
Nov	67.92	65.69	-3.29%	86.05	86.10
Dec	72.40	70.18	-3.06%	88.30	88.83

Finding:

Nine of twelve 2025 months carried fewer passengers than the same 2024 month. February had the largest relative decline (- 4.77%), followed by November (-3.29%) and December (-3.06%); October increased 1.71%. March through June had lower passengers alongside higher seat supply. This pattern helps choose months for a route-level review, but it does not by itself prove a seasonal or economic cause. Changes were calculated before rounding.

05 / FINDINGS - CARRIER CHANGE

Q3. Which airlines gained or lost the most passengers between 2024 and 2025?

The two tables show changes in passenger counts, in millions. Each table ranks reporting carriers by the size of their gain or loss.

Top 5 airlines with passenger gains

Code / reporting carrier	2024 pax (M)	2025 pax (M)	Gain (M)	2024 seats (M)	2025 seats (M)
UA United Air Lines Inc.	106.28	109.89	+3.61	126.96	134.85
OO SkyWest Airlines Inc.	40.59	44.10	+3.50	51.07	56.48
9E Endeavor Air Inc.	11.99	14.02	+2.02	14.68	17.61
YX Republic Airline	17.54	19.55	+2.01	22.24	25.19
MX Breeze Aviation Group	4.21	6.21	+2.00	5.60	8.25

Finding:

United gained 3.61 million passengers, the largest increase shown, with seats up from 126.96 to 134.85 million. SkyWest gained 3.50 million passengers as seats rose from 51.07 to 56.48 million. Endeavor, Republic and Breeze each gained about 2 million. These are reporting-carrier results, including regional operators, rather than a ranking of airline groups or consumer brands.

Top 5 airlines with passenger losses

Code / reporting carrier	2024 pax (M)	2025 pax (M)	Loss (M)	2024 seats (M)	2025 seats (M)
NK Spirit Air Lines	39.87	28.67	-11.21	48.63	36.99
WN Southwest Airlines Co.	173.18	165.83	-7.35	224.97	223.17
AA American Airlines Inc.	137.88	132.03	-5.85	164.81	162.09
DL Delta Air Lines Inc.	141.31	138.54	-2.78	165.46	167.13
ZW Air Wisconsin Airlines Corp	2.12	0.40	-1.72	2.58	0.52

Finding:

Spirit fell 11.21 million passengers, the largest decline, while its seats fell from 48.63 to 36.99 million. Southwest and American followed with losses of 7.35 and 5.85 million. Delta carried 2.78 million fewer passengers despite slightly more seats. The national decline was uneven; these differences warrant closer review of route mix and capacity before making a business explanation.

Values in millions are rounded to two decimal places after aggregation. Displayed deltas can differ by 0.01 from subtracting the rounded year columns.

06 / FINDINGS - DIRECTIONAL ROUTES**Q4. Which directional nonstop segments carried the most passengers in 2025?**

Each origin-to-destination direction is listed separately. Airport names accompany the codes for immediate interpretation.

Origin airport	Destination airport	Passengers
LAX - Los Angeles International Airport	JFK - John F. Kennedy International Airport	1,463,719
JFK - John F. Kennedy International Airport	LAX - Los Angeles International Airport	1,448,659
ORD - O'Hare International Airport	LGA - LaGuardia Airport	1,387,132
LGA - LaGuardia Airport	ORD - O'Hare International Airport	1,382,413
LAX - Los Angeles International Airport	SFO - San Francisco International Airport	1,365,176
ATL - Hartsfield-Jackson Atlanta International Airport	MCO - Orlando International Airport	1,357,393
SFO - San Francisco International Airport	LAX - Los Angeles International Airport	1,342,956
MCO - Orlando International Airport	ATL - Hartsfield-Jackson Atlanta International Airport	1,339,829
LAS - Harry Reid International Airport	LAX - Los Angeles International Airport	1,230,598
DEN - Denver International Airport	PHX - Phoenix Sky Harbor International Airport	1,198,989

Finding:

Los Angeles International Airport to John F. Kennedy International Airport led with 1,463,719 passengers; the reverse direction ranked second at 1,448,659, a difference of 15,060. O'Hare-LaGuardia and LaGuardia-O'Hare likewise appeared in both directions near the top. These pairs show substantial two-way segment activity, while Las Vegas to Los Angeles and Denver to Phoenix also reached the top ten.

A route here is one directional nonstop segment, not a complete itinerary. The table combines reporting carriers and aircraft types. A two-way corridor ranking, competitive position, or available capacity would need a separate calculation; passenger volume alone cannot establish route profitability.

07 / FINDINGS - CARRIER CAPACITY

Q5. Which reporting carriers supplied the most seats in 2025?

Code / reporting carrier	Seats (M)	Pax (M)	Pax / seats
WN Southwest Airlines Co.	223.17	165.83	74.31%
DL Delta Air Lines Inc.	167.13	138.54	82.89%
AA American Airlines Inc.	162.09	132.03	81.46%
UA United Air Lines Inc.	134.85	109.89	81.49%
OO SkyWest Airlines Inc.	56.48	44.10	78.08%
AS Alaska Airlines Inc.	41.16	33.05	80.28%
F9 Frontier Airlines Inc.	40.68	31.69	77.89%
NK Spirit Air Lines	36.99	28.67	77.50%
B6 JetBlue Airways	35.32	28.54	80.80%
YX Republic Airline	25.19	19.55	77.62%

Finding:

Southwest supplied 223.17 million seats, about 56.04 million more than second-ranked Delta. Southwest's 165.83 million passengers represented a 74.31% aggregate passenger-to-seat ratio, while Delta's was 82.89%. The table ranks seat supply; its ratios describe broad utilization across different route lengths and cannot explain profitability or service quality.

The ratio here is $100 \times \text{SUM}(\text{PASSENGERS}) / \text{SUM}(\text{SEATS})$. It is deliberately labeled passengers / seats because Q1's distance-weighted measure gives longer segments more weight. Compare carrier ratios cautiously when network mix differs.

08 / FINDINGS - DEPARTURE AIRPORTS

Q6. Which airports had the most departing passengers in 2025?

Origin airport	Departing passengers (M)
ATL - Hartsfield-Jackson Atlanta International Airport	44.23
DEN - Denver International Airport	37.69
DFW - Dallas Fort Worth International Airport	35.11
ORD - O'Hare International Airport	33.57
LAX - Los Angeles International Airport	24.89
LAS - Harry Reid International Airport	24.79
MCO - Orlando International Airport	23.99
PHX - Phoenix Sky Harbor International Airport	23.77
CLT - Charlotte Douglas International Airport	23.62
SEA - Seattle-Tacoma International Airport	22.03

Finding:

Hartsfield-Jackson Atlanta International Airport led with 44.23 million departing segment passengers, 6.54 million more than Denver International Airport's 37.69 million. Dallas Fort Worth and O'Hare followed at 35.11 and 33.57 million. The difference is a volume ranking in this scheduled U.S. carrier dataset, not a measure of unique trip origins or airport-wide international activity.

A connecting passenger may begin another segment at the same airport and contribute to the origin count. Airport planning should combine these figures with terminal throughput, transfer volumes, international service and operational capacity before drawing an infrastructure conclusion.

09 / EVIDENCE-BASED RECOMMENDATIONS

Prioritize the capacity-demand mismatch first. Each recommendation below begins with a measurable follow-up in the existing segment data; operational decisions would need the additional evidence listed alongside it.

Priority and evidence	Recommended next analysis	Before an operational decision
1 Capacity vs demand: 2025 pax -1.31%, seats +1.14%, weighted LF down 2.01 points	Compare 2024 and 2025 by carrier, route and month. Flag meaningful passenger declines with higher seats, then review utilization and departures.	Check fares/yield, schedule changes, competition and route economics before changing service.
2 Carrier shifts: Spirit -11.21M, United +3.61M passengers	Break the largest gains and losses into route and month contributions; compare seat changes on the same segments.	Confirm reporting-carrier relationships, network changes and schedules before attributing a cause.
3 Weaker months: nine of 12 fell; February -4.77%, November -3.29%, December -3.06%	Compare the same calendar months across the full five years, then inspect the leading carrier-route contributors.	Use a multi-year baseline, fares and schedules before treating a change as seasonal or altering capacity.

Supporting context

The LAX-to-JFK directional route and Atlanta origin rankings help select corridors and airports for that follow-up. High passenger volume alone is not a recommendation to add flights or expand facilities. Southwest's 223.17 million seats likewise identifies scale, while route mix and utilization determine where to investigate.

Data governance recommendation

Keep the service-class profile, annual row reconciliation, zero-passenger count and invalid-value checks as saved queries. Re-run the same validation when annual BTS files are refreshed. Record source download dates and schema changes so that a future comparison uses a consistent population.

10 / LIMITATIONS AND RESPONSIBLE INTERPRETATION

These are BTS monthly segment aggregates for class F scheduled U.S. carrier service after the documented filters. They support comparisons of observed segment volume and offered seats, with the following boundaries:

Boundary	Practical consequence
Segment passengers are not unique travelers	Connections can contribute multiple segment boardings.
Route direction matters	LAX to JFK and JFK to LAX are separate rows in the route ranking.
Carrier is the reporting carrier	Regional operators may appear apart from marketing brands or airline groups.
No fares, revenue or costs	Passenger totals and seat ratios are not profitability measures.
No cause variables	The report cannot attribute the 2025 dip to pricing, weather, economics or competition.
Filtered analysis population	Some raw class F records fail validity rules; zero passengers alone does not remove a row.
Airport origin is a segment origin	It does not uniquely identify where a complete passenger journey began.

Executive summary

The project documents a reproducible five-year BTS segment analysis and identifies where demand and capacity shifted. Its recommendations prioritize further investigation; the data do not establish causes, financial outcomes or individual passenger behavior.

11 / SQL SKILLS AND QUERY CATALOGUE

SQL skills demonstrated

Skill	Application in this project
Consistent schema	Loaded five yearly CSVs with the same explicit field types.
Source profiling	Grouped service classes and checked year/month coverage.
Conditional counting	Counted zero or invalid departures, seats, passengers, distance and months.
Reusable view	Centralized class, date and validity rules for all business questions.
Aggregation	Used SUM and GROUP BY for year, month, carrier, route and airport results.
Defensive division	Used SAFE_DIVIDE for percentages and utilization ratios.
Ranking and comparison	Sorted leading carriers, directional routes and origin airports.
Interpretation	Separated observed changes from hypotheses requiring more data.

Saved query catalogue

Query	Purpose
01 Service class profile	Establish class mix in the combined source table.
02 Scheduled quality check	Measure suspicious class F records, including zero passengers.
03 Create analysis view	Set a consistent scheduled segment population.
04 Annual demand/capacity	Answer Q1 and calculate distance-weighted load factor.
05 Monthly comparison	Answer Q2 for 2024 versus 2025.
06 Airline changes	Answer Q3; separate top gains and top losses in this report.
07 Top routes	Answer Q4 with directional 2025 airport pairs.
08 Carrier capacity	Answer Q5 with seats, passengers and aggregate ratio.
09 Departure airports	Answer Q6 with origin-airport passenger totals.

12 / DEFINITIONS, CONCLUSION AND SOURCE

How the measures were calculated

Measure	Calculation and interpretation
Passengers / seats / departures	SUM(PASSENGERS), SUM(SEATS), SUM(DEPARTURES_PERFORMED) across valid view rows.
Distance-weighted load factor	$100 \times \text{SUM}(\text{PASSENGERS} \times \text{DISTANCE}) / \text{SUM}(\text{SEATS} \times \text{DISTANCE})$. This compares passenger miles to available seat miles across segments; it is not a carrier's financial load factor.
Pax / seats	$100 \times \text{SUM}(\text{PASSENGERS}) / \text{SUM}(\text{SEATS})$. Used for Q5; each seat counts equally regardless of flight distance.
Year-over-year change	$100 \times (\text{2025 passengers} - \text{2024 passengers}) / \text{2024 passengers}$. Deltas and percentages are computed before presentation rounding.
Reporting carrier	UNIQUE_CARRIER and UNIQUE_CARRIER_NAME from BTS. These need not match marketing brands or corporate groups.

Conclusion

This case study shows a complete SQL workflow: loading comparable annual files, validating source values, defining a reusable analysis view, answering business questions and translating the results into bounded recommendations. The central finding is a 2025 passenger decline alongside continued growth in offered seats and departures. The monthly and carrier splits identify where to investigate, while the route and airport rankings give geographic and network context. Further fare, schedule and cost data are needed before explaining causes or deciding operational changes.

Reproducibility

BigQuery project: portfolio-project-508812; dataset: us_domestic_passenger_traffic; source table: t100_domestic_segments; analysis view: vw_scheduled_segments_2021_2025. The view applies the documented class, period and validity filters to all six business questions.

Official source documentation

BTS TranStats, T-100 Domestic Segment table profile and field descriptions.

BTS, T-100 segment data product reference (service class and field definitions).

The data are publicly downloadable from BTS TranStats. Cite BTS as the source; review the current source terms before redistributing the raw CSV files. This PDF reports derived aggregates and does not bundle the source data.